SILTHAUL · PRODUCT REQUIREMENTS AND SPECIFICATIONS

SiltHaul product requirements and specifications

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-11-03	Amish Chadha	Kit 1.9.1: migrated to the product requirements format (PR-001 to PR-012 are R1 to R12, wording unchanged), traced to the user requirements in SLH-URS-001; PR-013 to PR-018 added for the new user needs; status rerun on the detailed model (heaviest lift 22.2 kg, kit 407 L)
0.6	2026-10-03	Amish Chadha	Amish's round-2 decision 2A (SLH-DDR-004): R9 restated as "fits a small hatchback with the rear seats folded; heaviest part 25 kg or less" and accepted as met on paper
0.4	2026-10-03	Amish Chadha	Amish's requirement decisions (SLH-DDR-003); R1 restated (no lifting or carrying of mud; at least 0.5 times the bucket crew) and met on paper with a second box on the return leg; R9 met on paper with a flat-pack frame; status rerun from SLH-CAL-001 v0.3
0.3	2026-10-03	Amish Chadha	

			TRL 3 status from SLH-CAL-001 on the constructable design (SLH-DDR-002); R10 restated against the value-engineering target
0.2	2026-10-03	Amish Chadha	TRL 2 requirements with concept status; R11 overload limit and R12 load holding added (SLH-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

SiltHaul product requirements and specifications

Fourteen of the eighteen product requirements are met on paper or by design for option A, two are at risk (PR-001 by a hair and PR-006, the chain just under its factor of 5), two (PR-008, setup time, and PR-014, durability) can only be verified at TRL 4, and none is reported not met (SLH-CAL-001 v0.5, Table 3). PR-001 to PR-012 are the twelve requirements R1 to R12 that Amish accepted on 2026-10-03 and 2026-10-03 round 2 (SLH-DDR-003, SLH-DDR-004), with their wording unchanged; the "Was" column keeps the old numbers. PR-013 to PR-018 were added at the kit 1.9.1 upgrade so that every user need in SLH-URS-001 has a product requirement; they are proposals, awaiting Amish. "Met on paper" means shown by calculation, not by test. The statuses are for option A, the existing design; the other options are scored against the same needs in SLH-OPT-001.

> **Safety:** SiltHaul is a rope-and-anchor system under tension with a hand-cranked chain drive. PR-006, PR-007, PR-011, PR-012 and PR-016 are its safety requirements; they are checked on paper only and must be proved by proof-load tests before anyone uses the kit (SLH-BLD-001, section 6).

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001, UR-002	Performance	Mud moved per hour by a crew of four	No lifting or carrying of mud; output at least 0.5 times that of the same crew with shovels and buckets (estimate)	0.5	min	ratio to a bucket crew	Side-by-side timed trial on a test bed of wet silt (TRL 4)	Met on paper, at risk: 0.51 against 0.90 m3/h (ratio 0.57) with a 1.5 min fill; 0.497 if the loaders fill at a shoveller's 10 L/min (SLH-CAL-001 D6); loaders push mud in over the lip; nobody lifts or carries it	R1
PR-002	UR-003	Usability	Crank handle force at full box	150 N (34 lbf) or less per person with two people at the cranks	150	max	N per person	Spring balance at the handle during trial (TRL 4)	Met on paper: 63 N each at 1,000 N (126 N for one person); 61 N each at the 977 N stroke estimate	R2
PR-003	UR-002	Performance	Scraper box capacity	About 40 L (10 US gal) of wet mud per trip	40	nominal	L	Measured fill on test bed (TRL 4)	Met by design: 40.3 L in each of the two boxes at a 245 mm fill	R3
PR-004	UR-004	Physical architecture	Fits through a doorway	Box, tail block and ramp pass a 0.7 m (28 in) opening, with the return leg beside the box	700	max	mm	Fit test (TRL 4)	Met by design: each box 258 mm over the shackles; the boxes pass 42 mm apart and cross the ramp 26 mm inside its cheeks; ramp 646 mm	R4
PR-005	UR-005	Performance	Working length	Up to 15 m (49 ft) from tail sheave to capstan	15	min	m	Trial at full length (TRL 4)	Met by design: 12 m of travel on each drum half; capstan at least 6 m from the door	R5

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-006	UR-006	Safety	Rope and anchor strength	Minimum break strength at least 5 times the working pull, and at least 5 times the PR-011 limit for the rope and chain	5	min	factor	CalRig proof-load test of each anchor type (TRL 4)	At risk: rope 18 times working, 5.4 at the limit; chain 4.93 at the 3,000 N limit (5.03 at the 2,941 N the pin can release); fix proposed to Amish (SLH-STM-001 S5)	R6
PR-007	UR-006	Safety	Tail anchor independent of walls	Holds the PR-011 limit on both legs on a bare concrete floor with no wall contact	3000	min	N rope tension	Pull test on the anchored tail block (TRL 4)	Met on paper on a sound slab: 0.45 of recommended anchor loads	R7
PR-008	UR-007	Usability	Setup time	Three people set up in 20 min or less	20	max	min	Timed trial (TRL 4)	To verify at TRL 4: about 20 min estimated, at the limit (bolting the frame and shackling two boxes add 4 min)	R8
PR-009	UR-008	Packaging and transport	Portability	Kit fits a small hatchback with the rear seats folded; heaviest part 25 kg or less (25 kg is 55 lb)	25	max	kg	Weigh and pack test (TRL 4)	Met on paper, as restated by Amish (2A, round 2): heaviest lift 22.2 kg (drum with bearings); flat-pack frame, largest piece 890 x 814 x 83 mm; kit about 407 L packed against a 594 L small car boot	R9

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-010	UR-009	Physical architecture	Cost	Prototype parts at or under the USD 2,000 value-engineering target (a hypothetical control target)	2000	max	USD	Costed bill of materials	Met on paper, within the target: USD 1,055.60 (USD 944.40 under)	R10
PR-011	UR-006	Safety	Overload limit	Rope tension limited to 3,000 N whatever force the crew applies	3000	max	N	Shear pin release test on the capstan (TRL 4)	Met on paper: shear pin releases at 2,451 N (1,961 to 2,941 N)	R11
PR-012	UR-006	Safety	Holds when let go	The drum holds the load in either direction when the cranks are released	2	min	ratchets and pawls	Release test at working pull (TRL 4)	Met by design: two ratchet wheels and pawls on the drum shaft	R12
PR-013	UR-010	Materials	Made from stock	Every made part is cut, drilled, bent or welded from stock sections, sheet, bar and tube; bought parts are commercial catalogue items; no part needs machining beyond drilling and sawing	1	nominal	local workshop	Analysis of the BOM and the 245-part model (TRL 3); prototype build (TRL 4)	Met by design: 40 x 40 x 2 SHS, 219.1 x 3 tube, sheet and bar; UCP bearings, 08B-1 chain and sprockets, rope and fasteners bought	
PR-014	UR-011	Materials	Survives mud and wet	Steel parts painted or zinc plated; bearings sealed and greaseable;	1	nominal	coating and wash-down	Wash-down and inspection after a trial (TRL 4)	To verify at TRL 4: coatings and sealed UCP bearings in the BOM;	

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				all surfaces that touch mud can be hosed clean with no closed pockets					coating life not assessed	
PR-015	UR-012	Safety	Crank crew clear of the mud	The capstan stands outside the door, at least 6 m from the ramp roller, so the crank crew is out of the mud; loaders push mud into the box at floor level	6	min	m	Layout drawing SLH-DWG-002; timed trial (TRL 4)	Met by design: capstan at least 6 m from the ramp roller (fleet angle 1.41 degrees); loaders push mud in at floor level	
PR-016	UR-006	Safety	Moving parts guarded	The chain and both sprockets are closed by a fixed guard; ratchet wheels and pawls are on the outer face of the frame, clear of the crank crew	1	min	closed guard	Inspection of the prototype (TRL 4)	Met by design: closed 1.5 mm sheet chain guard (BOM 9) held by two M6 screws	
PR-017	UR-003	Usability	Tipping a full box	Two people on the tipping bar start a full box tipping with 150 N or less each	150	max	N per person	Spring balance at the bar (TRL 4)	Met on paper: 237 N to start a full box, 119 N each with two people (SLH-CAL-001 K3)	
PR-018	UR-013	Performance	No added water	The kit moves the mud as it is and adds no water, so the wet waste	1	max	m3 delivered per m3 of mud	Analysis (TRL 3)	Met by design for option A (no flush water); option C would deliver 2.8 m3	

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				leaving the room equals the mud removed					per m3 (SLH-OPT-001)	

Status words: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Requirements at risk or not met

None is reported not met for option A. Two are at risk and the rest wait for TRL 4:

- **PR-006 (R6).** Drum bearing friction raises the chain pull to 3,612 N at the 3,000 N limit, a factor of 4.93 on the 17.8 kN breaking load, just under 5; 5.03 at the 2,941 N the pin can release. The independent review found it. Three fixes are put to Amish (SLH-STM-001 S5): a 3.5 mm pin (limit about 2.25 kN), duplex 08B-2 chain, or accept it.
- **PR-001 (R1).** The ratio of 0.57 rests on estimated loading, dumping and bucket rates and on loaders filling 13.4 L/min each; at a shoveller's 10 L/min the ratio is 0.497 (SLH-CAL-001 D6). A timed trial settles it. The restatement is Amish's (2A, 2026-10-03): "no lifting or carrying of mud; output at least 0.5 times the same crew with shovels and buckets".
- **PR-008 (R8).** About 20 min with three people, at the limit, now that the frame is bolted together on site and two boxes are shackled.
- **PR-009 (R9).** Accepted by Amish with the rear seats folded (round 2, 2A, 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed."). With the seats up the 890 x 814 mm side frames do not lie flat in a typical small boot; that is no longer an open question.
- **PR-011 (R11).** The release of 2,451 N rests on a shear strength of 240 MPa and 20 % scatter; the release test at TRL 4 settles it.

For the other options (SLH-OPT-001): option B does not meet PR-002 (189 N each with four people) and has no PR-011 or PR-012 equivalent; option C does not meet PR-001 (0.49), PR-002 (183 N each with two people) or PR-018 (2.8 m3 per m3).

Assumptions

- Most flood mud is fluid enough to be scraped and dragged, not dug; very stiff or deep deposits may still need shovels first.
- The building has a sound concrete slab at least 100 mm thick for the tail block anchors; where it does not, SiltHaul is not used in that room.
- Volunteers accept a short briefing on rope safety.
- "A small hatchback with the rear seats folded" is taken as a small hatchback's load space with the rear seats folded: 1,250 mm long, 950 mm wide and 500 mm high (about 594 L).
- The detailed model of kit 1.9.1 (245 parts) changed the packed volume from 454 L to 407 L and the heaviest lift from 23.1 kg to 22.2 kg because the drum and fasteners are now modelled part by part; no design dimension changed.